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RBC Model with Staggered Nash Bargaining for Italy

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Chapter 1

Introduction

1.1 Purpose

The aim of this dissertation is to describe the Italian labor market with Gertler and Trigari (2009) model, a model that incorporates search and matching frictions, as well as staggered Nash bargaining. The model was specifically developed to answer one of the main failures of RBC modeling *i.e.* the fact that even the MP[1] could not account for the key cyclical movements of the U.S. labor market, for reasonable calibrations. After this puzzle was discovered, much of the literature started to find possible solutions, starting from a microeconomic explanation for wage rigidity[2]. Because of the complexity of this research of axiomatic foundations, difficulties in developing a quantitative model arose. In this framework Gertler and Trigari (2009), incorporated in a fully specified DSGE model in Gertler, Sala, and Trigari (2008), inserted wage rigidity in a more practical way. In particular, they introduced a sort of Poisson's adjustment, highly used in DSGE modeling in various contexts, through a staggered Nash bargaining. This approach turned out to be successful concerning moment matching. Worth of mention that, this is not the only solution proposed to solve the puzzle. Hagedorn and Manovskii (2006) consider an alternative parameterization to solve the business cycle puzzle. Because of the importance

[1] Mortensen and Pissarides

[2] Among the others Menzio(2005), Kennan (2006)

of the approach of Gertler and Trigari (2009), it seemed logical to apply the same model to Italy. For sure the possibility of using a DSGE arose to replicate Italian business cycle statistics, but because of the main purposes that was understanding the performance of the staggered Nash bargaining idea against the Italian labor market, we used an RBC. During the process of replicating Gertler and Trigari (2009) we considered the idea of augmenting the model with efficiency wages trying to incorporate staggered Nash bargaining with the ideas of some authors such as Vasilev (2021b) and Martin and Wang (2018), try that was unsuccessful as it will be pointed out in next section. Because of this we decided to judge the original model against Italian data. As a result, we found Gertler and Trigari (2009) can't account for some moments, namely the relative standard deviation of employed and unemployed. We suggest that this is due to the quarterly calibration of the model, rather than the monthly Gertler and Trigari used for the US, and this is because, ceteris paribus, the stickiness coefficient in a quarterly calibration is much lower. In the next section, we will explain why we could not merge staggered Nash bargaining and efficiency wages; in section (2), we report for clarity reasons the basics equations of the Gertler and Trigari (2009) model, in section (3) we will present the calibration and in section (4) we will point out the results.

1.2 Integrating efficiency and matching frictions

As we previously recalled, one possibility explored was to merge efficiency wages and staggered Nash bargaining. It turns out to be unsuccessful for two different reasons. In the first place, imposing that the Nash bargaining respect the Shapiro and Joseph E. Stiglitz (1984) condition is not a feasible option. Formally this means that, given that the utility of worker is $U(e) = c + (1 - e)\Gamma$ where w^r the reservation wage, $e \in \{0, 1\}$ is the effort, 1 if the effort is exerted, 0 if not and d is the detection rate, the salary should be at a level that makes optimal for the worker to exert effort. This means that:

$$\omega \geq (1-d)(w+\Gamma) + d\omega^r \Leftrightarrow \omega \geq \frac{1-d}{d}\Gamma + \omega^r \tag{1.1}$$

As we previously exposed this is not a feasible possibility, since, given a reasonable calibration of the parameters, the wage that results from the Nash bargaining is greater than the efficiency wage. This makes the constraint on the wage $\omega \geq \frac{1-d}{d}\Gamma + \omega^r$ not binding and always respected. Some authors, *e.g.* Walsh (2012), already showed that efficiency wages only affect weak bargaining power workers, such as non union workers, and for our calibration bargaining power was too high to make the constraint binding.

Another possibility explored was to create an incentive in the Nash bargaining due to the presence of gift exchange, following the idea of Arkelof: *"On the worker's side, the 'gift' given is work in excess of the minimum work standard; and on the firm's side the 'gift' given is wages in excess of what these women could receive if they left their current jobs" (Akerlof (1982)).* Formally this means that the effort is determined similarly to Danthine and Kurmann (2004). This created problems because of the presence of staggered Nash bargaining, to be precise only a fraction of firms renegotiate, and an effort similar to Danthine and Kurmann (2004) makes the effort vary on the basis of the gift perceived by the worker, that depends also on the difference between the salary and a reference salary. To set the reference salary one typical way it is to make the reference salary be an alternative salary, that in the framework of Gertler and Trigari (2009) would be a medium salary in the economy. There is the problem, anyway, that by having just a fraction of firms renegotiating the effort would vary during time, with the risk of being incompatible with Gertler and Trigari (2009) assumption of no voluntary separation, during work life, between workers and firms. Any attempt to solve the problem resulted in a function that was not *well behaved*, making the simulation impossible. Anyway this line of reasoning is fascinating and we hope, in the future, to arrive to a solution.

A third way to insert efficiency wages was to follow Vasilev (2021b) and Martin and

Wang (2018), where efficiency wages and search and matching frictions are present, but in their model, wages are set by the firm, and there is no bargaining. This was away from the aim of integrating staggered Nash bargaining and efficiency wages, furthermore, there are difficulties in making coexist a matching function and the fact that the firm imposes the wage. In a matching model, in particular, the frictions between demand and supply of work create a local monopoly rent that is shared, resulting in a decentralized firm worker bargaining.

Even with these failures, efficiency wages, are, from a theoretical point of view, a very powerful concept and we believe that their introduction would increase the stickiness of the wages, we hope in the future to solve the functional problem that prevented their introduction.

Chapter 2

RBC model

In this section, we will develop the model of Gertler and Trigari (2009) by producing a labor market RBC with staggered Nash wage bargaining. We will work in the trace of Mortensen and Pissarides under the assumption of Hansen (1985) and Rogerson (1988) of indivisible labor supply. Recursive methods are used to solve the dynamic problems that will arise and log linearization of the model will follow.

2.1 Labor market and the matching function

There is a continuum of workers of measure 1, and the workers are either employed or unemployed. By aggregating the data we obtain that n_t is the number of employed at time t and $u_t = 1 - n_t$ is the number of unemployed at time t. Each worker employed obtains the wage $w_t(i)$ that he bargained with the employer while unemployed workers search for work. Each firm posts vacancies to attract workers, and those vacancies will, eventually, be filled in the next period. The total number of vacancies posted at the time t is v_t. As we said, vacancies will *eventually* be filled. Some workers are searching and firms post vacancies, but there is no guarantee that a vacancy will be filled in the next period. What we are saying is that there is a need for some sort of technology, a *matching function*, that matches vacancies with workers. The matching function, as Andolfatto (1996) explains, it is assumed to be non - decreasing, concave and with CRS. The function's input are usually vacancies and aggregate effort $e(1-n_t)$. Since in our model as in Gertler and Trigari (2009) we disregard the direct search of workers, the matching function can be expressed as $M_t = f(v_t; (1-n_t))$. According to Gertler and Trigari (2009) we set the matching function to be:

$$m_t = \sigma_m u_t^{\sigma} v_t^{\sigma} \tag{2.1}$$

Where the probability a firm fills a vacancy q_t and the probability that an unemployed finds work s_t are defined identically to Gertler and Trigari (2009):

$$q_t = \frac{m_t}{v_t} \tag{2.2}$$

$$s_t = \frac{m_t}{u_t} \tag{2.3}$$

We also assume, again following Gertler and Trigari (2009), that $(1-\rho)$ workers and firms are separated exogenously, so the probability of keeping the work in the next period from the perspective of a worker, is ρ. This means that the fluctuations in unemployment are connected to hiring and not from the separation of workers and firms, precisely as in our reference. We should also find a law of motion for employment. We can express the law of motion of employment as:

$$n^{'} = \rho\, n + m \; with \; \rho \in [0,1] \tag{2.4}$$

2.2 Firms

Each firm produces with a Cobb - Douglas technology; the input used to produce output $y_t(i)$ are capital $k_t(i)$, labor $n_t(i)$. Since $\bar{h}$ is equal between workers and is exogenously set we won't consider this input in the production function:

$$y_t(i) = a_t k_t(i)^{\alpha} n_t(i)^{1-\alpha} \tag{2.5}$$

Where a_t is a common productivity factor that evolves according to a Markov process:

$$ln(a_t) = \bar{a} + \tilde{a}_t \tag{2.6}$$

Where the random component evolves according to:

$$\tilde{a}_t = \rho_a \tilde{a}_{t-1} + \epsilon_t^a \tag{2.7}$$

Note that in this model, no friction or capital control exists as well as we have a competitive rental market in capital. Note that this is a standard assumption in standard RBC models in line with Kydland and Prescott (1982). New DSGE models include frictions in the capital markets, but with the intention of studying the labor market in a RBC environment, we won't follow this approach. As in Gertler and Trigari (2009) we define the hiring rate in period t as:

$$x_t(i) = \frac{q_t v_t(i)}{n_t(i)} \tag{2.8}$$

Equation (2.4) can be rewritten for a single firm as:

$$n^{'}(i) = \rho n(i) + q_t v_t(i) \tag{2.9}$$

We follow again Gertler and Trigari (2009) rather than the standard search and friction literature (*e.g.* Diamond–Mortensen–Pissarides (DMP) labor search model) in imposing a quadratic adjustment labor cost in posting a vacancy rather than a fixed cost for posting a vacancy. This is a forced choice we need to follow if we want to follow their approach of staggered Nash bargaining, otherwise in the presence of wage dispersion and non - quadratic adjustment costs, no determined equilibrium is ensured. As we will always do we will find the problem F.O.C. by using a recursive approach. A detailed introduction is present in Appendix A, where the fixed point theorem, metric spaces and Bellman equation as an application of the fixed point theorem are explained. It has to be noted that in setting up the recursive problem, we need to take into account that we can not discount tomorrow's value function with just β that is the subjective discount factor for households. In particular, β can be used to discount utilities not profits, to discount profits we should also consider the inter - temporal substitution between today's and tomorrow's consumption. Define $\Lambda_{t,t+1} = u_{c'}/u_c$ then the discounting factor for the firm is $\beta E_t \Lambda_{t,t+1}$. The firm chooses vacancies and capital to maximize the discounted sum of profits, and the firm does not choose neither the wages neither the effort level; the effort level will be considered in the Nash bargaining when the surplus of an additional worker will be computed, in particular the firm will have to pose attention to the bargaining process considering that the amount of wage that will be set will influence the firm profits. The problem can be stated in recursive terms as follows:

$$V(n;a) = \max_{k,v}\{a_t k_t(i)^\alpha (n_t(i))^{1-\alpha} - wn - rk - \frac{\kappa}{2} x_t(i)^2 n_t(i) + \beta E \Lambda_{t,t+1}\left[V(n^{'};a^{'})\right]\} \quad (2.10)$$

$$n^{'}(i) = \rho n(i) + q_t v_t(i) \quad (2.11)$$

Deferring the bargaining process F.O.C. for capital and vacancies can be computed as follows, derive (2.10) for both capital and vacancies and impose that the derivatives are equal to zero, starting from capital:

$$a\alpha\left(\frac{k(i)}{n(i)}\right)^{\alpha-1} - r = 0 \quad (2.12)$$

Rewriting (2.12) we can obtain the classical inter - temporal solution.

$$r = a\alpha\left(\frac{k(i)}{n(i)}\right)^{\alpha-1} \quad (2.13)$$

Lets now work on the vacancy posting, and the problem can be expressed as a Lagrangian:

$$\begin{aligned} F(n^{'}, v, \lambda) = a_t k_t(i)^\alpha n_t(i))^{1-\alpha} - wn - rk - \frac{\kappa}{2} x_t(i)^2 n_t(i) + \beta E \Lambda_{t,t+1}\left[V(n^{'};a^{'})\right] + \\ \lambda\{\rho n(i) + q_t v_t(i) - n^{'}(i)\} \end{aligned} \quad (2.14)$$

From eq. (2.14) we obtain that, imposing the conditions for the maximum:

$$\begin{cases} -\kappa x_t(i) q_t + \lambda q_t = 0 \\ \lambda = \beta \Lambda_{t,t+1} V_{n'}(n^{'}; a^{'}) \end{cases} \tag{2.15}$$

From Benveniste and Scheinkman (1979) and the envelope theorem[1] :

$$V_n(n; a) = (1-\alpha)\frac{y_t(i)}{[n_t(i)]} - \frac{\kappa}{2} x_t(i)^2 + \lambda\rho \tag{2.16}$$

From the second equation in (2.15) and (2.16) we obtain that:

$$V_n(n; a) = (1-\alpha)\frac{y_t(i)}{[n_t(i)]} - \frac{\kappa}{2} x_t(i)^2 + \rho\beta\Lambda_{t,t+1} V_{n'}(n^{'}; a^{'}) \tag{2.17}$$

Lagging of one period we obtain:

$$V_{n'}(n^{'}; a^{'}) = (1-\alpha)\frac{y^{'}(i)}{n(i)^{'}} - \frac{\kappa}{2} x^{'}(i)^2 + \rho\beta\Lambda_{t+1,t+2} V_{n''}(n^{''}; a^{''}) \tag{2.18}$$

So we can condense the system (2.15) as follows:

$$kx(i) = \beta\Lambda_{t,t+1} V_{n'}(n^{'}; a^{'}) \tag{2.19}$$

Where $V_{n'}(n^{'}; a^{'})$ has been defined in (2.18). What has been derived in eq. (2.19) is pretty straightforward, a firm will choose to post vacancies until the point in which the marginal cost of posting vacancies, left - hand side of eq. (2.19) is equal to the marginal benefit of having an employee, right hand side of eq. (2.19). To sum up the results of this section, we obtained F.O.C. for firms that maximize the discounted value of profits (*i.e. firms maximize their value*) and we obtained two classical inter temporal conditions: (2.13), (2.19)

[1] Both theorems are presented in the appendix.

2.3 Households

It is possible, by using lotteries, to show that individuals are better off if they pool resources together. This can easily be proven by noting the concavity of the logarithm. Having aggregated the individuals in the multi-member household construct, we can now tackle consumption-saving decisions; the problem can be stated as follows:

$$U = \max_{c}\{ln(c) + \beta EU^{'}\} \tag{2.20}$$

$$k^{'} = (1 + r - \delta)k + wn + (1 - n)b + \Pi - T - c \tag{2.21}$$

We can substitute in equation (2.20) consumption with the constraint then the the problem becomes:

$$U = \max_{k^{'}}\{ln((1 + r - \delta)k - k^{'} + wn + (1 - n)b + \Pi - T) + \beta EU^{'} \tag{2.22}$$

Imposing optimality condition the F.O.C. becomes:

$$-\frac{1}{((1 + r - \delta)k - k' + wn + (1 - n)b + \Pi - T)} + \beta EU^{'}_{k'} = 0 \tag{2.23}$$

Noting that at the denominator there is c we substitute and then we note (without using any theorems on the derivative of the value function) that:

$$U^{'} = \max_{k^{''}}\{ln((1 + r^{'} - \delta)k^{'} - k^{''} + w^{'}n^{'} + (1 - n)^{'}b^{'} + \Pi^{'} - T^{'}) + \beta EU^{''} \tag{2.24}$$

We can find that:

$$U^{'}_{k} = \frac{(1 + r^{'} - \delta)}{c^{'}} \tag{2.25}$$

By combining equation (2.23) with equation (2.25), we obtain the inter-temporal decision between today and tomorrow's consumption:

$$\frac{1}{c_t} = \beta E_t \frac{(1 + r^{'} - \delta)}{c_{t+1}} \tag{2.26}$$

Equation (2.26) is the famous Euler equation for consumption.

2.4 Wage setting

This section will describe Gertler and Trigari (2009) labor market. This is the fundamental feature of the model; rather than a simple period-by-period bargaining between workers and firms, only a fraction $(1-\lambda)$ firms will renegotiate the contract with all the workforce. In this case λ works as a degree of stickiness in a Poisson's adjustment. Due to this staggered bargaining, the average wage will be sticky and will depend on previous time average salaries.

2.4.1 Workers

We define as in Gertler and Trigari (2009) $V_t(i)$ as the value of employment for a worker. Each period a worker suffer a probability $(1 - \rho)$ of losing the job, since we imposed in (2.1) an exogenous separation rate of $(1 - \rho)$. This means that in recursive terms the value of employment is:

$$V(i) = w(i) + \beta E_t \Lambda_{t,t+1} \left[\rho V_{t+1} + (1 - \rho) U_{t+1} \right] \tag{2.27}$$

Let's now turn the attention to the value of unemployment that right now is not defined. The value of unemployment, heuristically, should be the sum of unemployment benefits, b, and the discounted (clearly considering the inter-temporal substitution between today and tomorrow consumption) expected value between the value of being a new hire in the next period (so with probability s_t, that remember is the ratio between matches at time t and unemployed) and the following period value function. So, formally:

$$U = b + \beta E_t \Lambda_{t,t+1} \left[s_t V_{x,t+1} + (1 - s_t) U^{'} \right] \tag{2.28}$$

Where $V_{x,t+1}$ is the the value of being a new hire in the new period. To compute the value of being a new hire in time $t+1$ consider that there is a probability $p(i)$ of being employed at firm i. If the worker is employed at firm i he receives $V(i)$ as defined in equation (2.27). The probability $p(i)$ is equal to the matches at firm i divided by the total matches in the economy. With some manipulation we now find that the probability is $p(i) = \frac{x_{t-1}(i)n_{t-1}(i)}{x_{t-1}n_{t-1}}$. Let's just compute the expected value and so we can define, as in Gertler and Trigari (2009) the value of being a new hire as:

$$V_{x,t+1} = \int_0^1 p(i)V(i)di \tag{2.29}$$

We can now compute worker's surplus, given by the difference between the value of employment and the value of unemployment, so subtracting equation (2.28) from equation (2.27) we obtain the surplus $H_t(i)$:

$$H_t(i) = w_t(i) - b + \beta E_t \Lambda_{t,t+1} \left[\rho H_{t+1}(i) - s_t H_{x,t+1} \right] \tag{2.30}$$

Where $H_{x,t} = V_{x,t} - U_t$.

2.4.2 Playful digression

The following few lines present a slight digression on the bargaining problem. In Gertler and Trigari (2009) model it will be applied the Nash bargaining rule, which is an axiomatic approach to bargaining; worth of mention axiomatic approach is not the only approach that is present in the literature and coexists with a complementary approach, namely the strategic approach. While the strategic approach describes the strategic conditions to reach an agreement in a non-cooperative game, the axiomatic approach initiated by Nash

(1950) defines a class of solutions consistent with some behavioural rules. In a matching model the frictions between demand and supply (described by eq. (2.1)) create a local monopoly rent that is shared, resulting in a decentralized firm worker bargaining. Let's now describe the feasible set, the set is a two-dimensional space of expected utilities of two agents (namely the firm and the worker) that we can easily describe with a disagreement point $d = (Z, U)$ where Z is the value of continued search for the firm and U is unemployment for the worker and with a set of feasible allocations in case an agreement is reached. Defining with ω_f and ω_w the reservation salaries for the firm and the worker then, in any agreement, it must be that $\omega_f \geq w \geq \omega_w$. The set of feasible allocation is, as ref note:

$$\Omega = \{x \in R^2 | \exists w \in R, \omega_f \geq w \geq \omega_w, x \preceq X(w), X(w) \succeq d\} \tag{2.31}$$

where $X(w)\ for\ \omega_f \geq w \geq \omega_w$ is the Pareto frontier. The couple (Ω, d), as we mentioned before, describes the bargaining problem. In our case, it is possible to show, but for clarity we will elude this explanation, that Ω is convex and compact and that the rational utility allocation is bounded. Under these assumptions the Nash bargaining solution is the argmax of the product of surplus of the worker and the firm, defining the surpluses respectively as H(i) and V(i):

$$w^* = arg \max_w H(i)V(i) \tag{2.32}$$

Where eq. (2.32) respects the behavioral properties of Pareto optimality, symmetry, invariance to utility representation and independence of irrelevant alternatives. A straightforward extension of eq. (2.32) to account for different power in the negotiation is:

$$w^* = arg \max_w H(i)^{\eta} V(i)^{1-\eta} \tag{2.33}$$

Equation (2.33) will be used in solving the wage problem.

2.4.3 Nash bargaining

As in Gertler and Trigari (2009) we restrict the possible contracts to contracts fixed on a set of time and we assume that each period firms have a $(1-\lambda)$ probability of renegotiating all the contracts. The average duration before renegotiating is clearly $1/(1-\lambda)$. As in our reference constant returns allow for the fact that at margin all workers are the same. As we introduced, wages will be set by a Nash-type negotiation and since both workers and firms are better off if they arrive at some sort of agreement (disagreement point is not a possible outcome of the negotiation) then, for a renegotiating firm, salaries are set by this maximization:

$$\max_{w_t^*} V_{n,t}(r)^{1-\eta} H_t(r)^{\eta} \tag{2.34}$$

Where $H_t(r)$ is the surplus of the worker and $V_{n,t}(r)$ is the surplus of the firm for an additional worker. Remember that we already defined both $H_t(r)$ in (2.30) and $V_{n,t}(r)$ in (2.17). Solution to the Nash bargaining is clearly given by this first order condition:

$$\frac{\delta V_{n,t}(r)}{\delta w_t^*} V_{n,t}(r)^{-\eta}(1-\eta) H_t(r)^{\eta} = -\frac{\delta H_t(r)}{\delta w_t^*}(\eta) H_t(r)^{\eta-1} V_{n,t}(r)^{1-\eta}$$

$$\frac{\delta V_{n,t}(r)}{\delta w_t^*} V_{n,t}(r)^{-\eta}(1-\eta) H_t(r)^{\eta} = -\frac{\delta H_t(r)}{\delta w_t^*}(\eta) H_t(r)^{\eta} H_t(r)^{-1} V_{n,t}(r)^{1} V_{n,t}(r)^{-\eta}$$

By simplifying and then manipulating we obtain that the F.O.C. can be expressed as:

$$\frac{\delta H_t(r)}{\delta w_t^*}(\eta) V_{n,t}(r) = -\frac{\delta V_{n,t}(r)}{\delta w_t^*} H_t(r)(1-\eta) \tag{2.35}$$

As in Gertler and Trigari (2009) we will try to find W_t^f and W_t^w that are the discounted value of paid wages for the firms and the worker. We will do so because the value of a new worker and the value of employment clearly depends not just on today's wage, but on the path of wages (that means on the negotiations that can happen with probability λ and on the fact that the worker keeps the job with probability ρ). So let's start with computing W_t^f, we can express it as:

$$W_t^f(r) = \Sigma_t(r)w_t^* + (1-\lambda)E_t\sum_{s=1}^{\infty}\frac{n_{t+s}}{n_t}(r)\beta^s\Lambda_{t,t+s}\Sigma_{t+s}(r)w_{t+s}^* \tag{2.36}$$

With $\Sigma_t(r)$ defined as:

$$\Sigma_t(r) = E_t\Sigma_{s=0}^{\infty}\frac{n_{t+s}}{n_t}(r)(\lambda\beta)^s\Lambda_{t,t+s} \tag{2.37}$$

Now computing $W_t(r)^w$ we find that:

$$W_t^w(r) = \Delta_t w_t^* + (1-\lambda)E_t\sum_{s=1}^{\infty}(\rho\beta)^s\Lambda_{t,t+s}\Delta_{t+s}w_{t+s}^* \tag{2.38}$$

With Δ_t defined as:

$$\Delta_t = E_t\sum_{s=0}^{\infty}(\rho\lambda\beta)^s\Lambda_{t,t+s} \tag{2.39}$$

We can now rewrite $J_t(r)$ and $H_t(r)$ as in Gertler and Trigari (2009) as:

$$J_t(r) = E_t\sum_{s=0}^{\infty}\frac{n_{t+s}}{n_t}(r)\beta^s\Lambda_{t,t+s}\left[f_{nt+s} - \frac{\kappa}{2}x_{t+s}(r)\right]^2 - W_t^f(r) \tag{2.40}$$

$$H_t(r) = W_t^w - E_t\sum_{s=0}^{\infty}(\rho\beta)^s\Lambda_{t,t+s}\left[b + s_{t+s}\beta\Lambda_{t+s,t+s+1}H_{x,t+s+1}\right] \tag{2.41}$$

By employing (2.35) we obtain that:

$$\eta\Delta_t = (1-\eta)\Sigma_t(r)H_t(r) \tag{2.42}$$

Equation (2.42) solves the bargaining problem.

2.5 Resource constraint

We end the description of the model with the resource constraint that Gertler and Trigari (2009), because of the presence of quadratic adjustment costs, expressed as:

$$y_t = c_t + k_{t+1} - (1-\delta)k_t + \frac{\kappa}{2}\int_0^1 x_t(i)^2 n_t(i)di \tag{2.43}$$

This ends with Gertler and Trigari (2009) model description; we will now assess how the model performs against Italian data.

Chapter 3

Calibration

We opted to calibrate the model rather than using bayesian inference to set the values of all the parameters. Calibration has been the classical way to set *RBC* models and can be thought as an estimation with a dogmatic point prior, *i.e.* there is no possibility for the likelihood function to update the mapping from prior to the posterior[1]. In more precise terms, we will assume some certain equivalence that will pin down the steady state of the model, and then the log - linearized model will be simulated in dynare, which use Schmitt-Grohe and Uribe (2004) method that is similar to Klein (2000) to solve the model[2]. In the Gertler and Trigari (2009) model there are 13 parameters that need to be assigned to a value. Namely five are conventional values: $[\beta,\ \alpha,\ \delta,\ \rho_a\ \sigma_a]$ where β is the discount factor, α is the capital share in the Cobb - Douglas production function, δ is the depreciation rate, ρ_a is the autoregressive parameter for the technology shock and σ_a is the standard deviation of the . The additional seven parameters that we need to calibrate are $\left[\rho,\ \sigma,\ \eta,\ s,\ \bar{b},\ \sigma_m,\ \lambda\right]$ where ρ is the survival rate, σ the matching function parameter, η the bargaining power parameter, s the steady state job finding probability, $\bar{b}$ the steady state flow contribution of the worker to the match[3], σ_m matching efficiency

[1]Some of these themes are presented in Pfeifer (2014).

[2]Other solution methods exists, among the others: Blanchard and Kahn (1980), Anderson (1997), Uhlig (1999). A comparison of different solutions methods is present in Anderson (2008).

[3]Namely determines the unemployment flow value b. To be clear:

$$b = \frac{\bar{b}}{f_n + \frac{\kappa}{2}x^2} \tag{3.1}$$

and λ average time before renegotiation. Because of the fact that the model will take into account Italy, and much of the existent literature is on the US post-war, we have small literature to obtain classical parameters. Therefore, we won't take standard parameters, instead we will compute many of the parameters. This will for sure reduce comparability among models, but should improve the performance of the model in replicating business cycle statistics. Among others Maffezzoli (2001) calibrates a non-Walsarian *RBC* for Italy; we will follow a similar approach to calibrate the parameters.

3.1 Standard parameters calibration

Let's now focus on calibrating the parameter δ, the depreciation rate. We know that the law of motion for capital is:

$$k^{'} = (1-\delta)k + I \tag{3.2}$$

Now, if we look at the steady state relationship we see that:

$$k_{ss} = (1-\delta)k_{ss} + I_{ss}$$

$$\delta = \frac{I_{ss}}{k_{ss}} \tag{3.3}$$

Equation (3.3) is what we use to calibrate the depreciation rate[4]. ISTAT contains series on *gross* capital and on investment, both for the non-farm sector. In the analysis we find an implied steady state of 3.61%, which implies an annual depreciation rate of the same value and a quarterly depreciation rate of 0.009 $\approx$ 1%. The value is compatible with Maffezzoli (2001) who finds a quarterly depreciation of 1.05% and with Marino (2016) who finds a depreciation of 1%.

For β we follow Maffezzoli (2001), Annicchiarico et al. (2013) and classical literature to set it at 0.99. Note that $\beta = 0.99$ quarterly is a standard parameter for *RBC* modeling;

In the dynare code we will directly calibrate b.

[4]It is also possible to use micro data on depreciation, we disregard this approach because, even though it is theoretically possible, complications arise during aggregation.

Gertler and Trigari (2009) use $\beta = 0.99^{1/3}$ because they want a monthly calibration[5]. However, as highlighted by Gertler and Trigari (2009), α in the model can't be interpreted as in the frictionless *RBC* as the labor share. This is somewhat tricky since, α remains the capital share, but data on capital share are typically more difficult to find. It's trivial to show, and it's a typical result for long-run macroeconomics, that α is the capital share *i.e.* the capital share it's only determined by the technology:

$$\max_k aK^\alpha L^{1-\alpha} - wn - kr \tag{3.5}$$

We can easily derive the F.O.C. of the problem:

$$\alpha z\left(\frac{L}{k}\right)^{1-\alpha} - r = 0$$
$$\alpha\frac{y}{k} - r = 0$$

That is indeed:

$$\alpha = r\left(\frac{k}{y}\right) \tag{3.6}$$

We set the capital share to $\alpha = 0.33$ following Annicchiarico et al. (2013). Before tackling ρ_a, the autoregressive parameter, we need to find an empirical measure for TFP. Then this measure will be linearly detrended, since in the model it is implicitly assumed that a linear deterministic trend drives the observed trends in the actual data[6]. To do so we will regress the empirical measure of total factor productivity on a constant and a linear trend:

$$ln(\hat{a}_t) = \phi_0 + \phi_1 t_1 + z_t \tag{3.7}$$

Then the measured residuals, *i.e* the detrended TFP series, are taken and by estimating

[5] Calibration on this specific parameter is possible by noting that:

$$\beta = \frac{1}{r_{ss} + 1 - \delta} \tag{3.4}$$

.

[6] See Sims (2015)

the AR(1) process:

$$\hat{z}_t = \rho_a \hat{z}_{t-1} + \varepsilon_t \tag{3.8}$$

we get the autoregressive parameter. To find a quarterly measure of TFP we can take logs of the Cobb - Douglas production function and note that:

$$Y = a_t k^{\alpha} n^{1-\alpha}$$

$$ln\ y_t = ln\ a_t + \alpha\ ln\ k_t + (1-\alpha)\ ln\ n_t$$

$$ln\ a_t = ln\ y_t - \alpha\ ln\ k_t - (1-\alpha)\ ln\ n_t \tag{3.9}$$

Note that to use eq.(3.9) we need a quarterly series of capital that ISTAT and other data sources, do not provide. To solve this problem, we use a variation of the permanent inventory approach, a classical solution for this kind of problem in literature[7]. This means that, having the initial physical stock of capital and the quarterly gross investment, we can numerically solve the quarterly depreciation that generates the next year stock of capital. Having a starting point and the quarterly investment and depreciation it is trivial to derive the quarterly series. To solve for each quarter depreciation rate, given $K^{'}$, end of the period stock K, beginning of the period stock and I_i, investment in quarter i:

$$K' = \sum_{t=1}^{4} \Big[(1-\delta)^i I_{4-i}\Big] + (1-\delta)^4 K \tag{3.10}$$

By solving numerically (3.10) we obtain a quarterly depreciation rate[8] for each quarter. Now we can simply use eq. (3.9), then we detrend the TFP and by estimating the AR(1) process on the residuals following the approach proposed in Sims (2015) we obtain that $\rho_a = 0.942$ and $\sigma_a = 0.62\%$. Note that we find a much higher autoregressive parameter

[7] Among the others also Maffezzoli (2001) use the same procedure to overcome the problem.

[8] There is a slight variation on the depreciation rate, anyway even though there is some variation, data are in line with calibration of δ.

than Maffezzoli (2001) that finds a autoregressive parameter of $\rho_a = 0.87$, while σ_a is really similar: (0.62% vs 0.56%). Results are summarized in *Table 1*.

Table 1: Autoregressive process

Autoregressive parameter	ρ_a	0.943***
Technology s.t.d.	σ_a	0.62%
R squared	R^2	0.87%

3.2 Model specific parameters calibration

We set ρ by recalling that $\frac{1}{1-\rho}$ is equal to the average contract length, as noted by Gertler and Trigari (2009). In *Employment, Skills and Productivity in Italy* a research project in partnership with JPMorgan Chase Foundation, Jérome Adda and Antonella Trigari[9], by looking at INPS[10]data for the period $2005-2015$ find a job separation rate of 0.0235, implying an employment spell of 42 months, that is equal to 12 quarters and implies a $\rho = 0.9167$. In the same research Jérome Adda and Antonella Trigari find that the unemployment spell is 11 months, which implies a job finding probability (quarterly) of $s = 0.36$, far away from the 0.5 monthly probability that Gertler and Trigari (2009) found for the US, this difference anyway comes as no surprise given the different flexibility of the two labour markets. Note that these data are compatible with transition probabilities for Italy and Mediterranean Europe. We also set $\sigma = 0.5$, according to literature[11] and Gertler and Trigari (2009). Note that also other search and frictions model for Italy use a similar value *e.g.* Schubert (2021). We set $\eta = 0.5$ so that this choice ensures that the Nash bargaining is symmetrical[12]. We normalize the efficiency of the matching process, as in the original paper, so $\sigma_m = 1$. In the model $\bar{b}$ is the relative unemployment flow value, Gertler and Trigari (2009) follow Shimer (2005) and impose $\bar{b} = 0.5$. We will follow a slightly different approach, they note that with that particular $\bar{b}$ they obtain a

[9] *The ins and outs of non- employment in Italy*
[10] Istituto Nazionale di Previdenza Sociale, National Social Security Agency
[11] Since no data is available for this parameter, an extensive sensibility analysis will be conducted.
[12] Which is the classical solution to Nash bargaining, Nash (1950).

replacement ratio of 0.42[13], so we will calibrate $\bar{b}$ to match the Italian replacement ratio $\varrho = 0.55$[14].

$$\varrho = \frac{\bar{b}}{w_{ss}/f_n} \tag{3.11}$$

We set $\bar{b} = 0.518$ and we can now note that κ is pinned down by[15]:

$$\begin{cases} \kappa x = \beta(f_n - w + \frac{\kappa}{2}x^2 + \rho\kappa x) \\ w = \chi(f_n + \frac{\kappa}{2}x^2 + s\kappa x) + (1-\chi)\bar{b}(f_n + \frac{\kappa}{2}x^2) \end{cases} \tag{3.12}$$

And respectively

$$b = \bar{b}(f_n + \frac{\kappa}{2}x^2) \tag{3.13}$$

So by solving the system we obtain implied values of $\kappa = 34.84$ and $b = 1.44$.

We also need to calibrate λ, also in this case we can notice that $\frac{1}{1-\lambda}$ is the average time before a renegotiation happens. Even though no direct evidence exists, Gertler and Trigari (2009) argue that renegotiations can happen between once per year as pointed by Taylor (1999), for unionized and ununionized workers, and far less than one year, because of adjustments such as bonuses. They choose a $\lambda = 8/9$ that in their monthly model means that the average time before a renegotiation occurs is 3 quarters. They also consider a 4 quarter renegotiation time, but just as a robustness exercise. Even though we share the idea that bonuses and other parts of the compensation scheme are on average adjusted more frequently than once a year; since Italian contracts are usually automatically revised once a year, and usually just on the basis of a predetermined parameter, we decided to increase the average time before a renegotiation happens to 4 quarters. This means that $\lambda = 0.75$. A summary of the calibrated parameters is present in *Table 2*.

[13] In the interpretation that b are unemployment benefits.

[14] OECD statistics, for average salary and unemployment duration equal to 11 months, equally to the one chose in calibration of s.

[15] To check the pin down of κ and b, we verified in the original paper and we obtained $\kappa = 148.2$ and $b = 1.46$

Table 2: Calibrated parameters

		IT q	**US m**
Production function parameter	α	0.33	0.33
Discount factor	β	0.99	0.997
Capital depreciation rate	δ	0.01	0.008
Technology autoregressive parameter	ρ_a	0.943	0.983
Technology standard deviation	σ_a	0.0062	0.0075
Survival rate	ρ	0.917	0.965
Elasticity of matches to unemployment	σ	0.5	0.5
Job finding probability	s	0.36	0.45
Bargaining power parameter	η	0.5	0.5
Adjustment cost parameter	κ	34.84	148.2
Relative unemployment flow value	$\bar{b}$	0.518	0.4
Unemployment flow value	b	1.44	1.46
Renegotiation frequency	λ	0.75	0.889
Efficiency of the matching process	σ_m	1	1

Chapter 4

Results

4.1 Data treatment

4.1.1 Business cycle treatment

The judgement of the model is against Italian data from $[2004:1-2019:4]$ normalized for working age population[1], for series available monthly we take quarterly averages. We decided to exclude the covid crisis because of the specific nature of the shock, an element that an *RBC* model with *TFP* shocks can't account for. The data are prevalently from ISTAT; output y represents the non farm production, n are employed in non farm sector. Unemployment u is equal to unemployed over 16 years old. All the data are logged and then filtered with the *Hodrick Prescott filter* with a conventional smoothing parameter to extract the cyclical component of the logged variables. The filter choice is a crucial point in any real business cycle analysis, and the filter choice, *i.e.* detrending, is the core of the analysis in the sense that different filters will give and highlight different business cycles. Even though no consensus exists, and numerous researchers *e.g.* Hamilton (2018) expressed high criticism, we went for a *H.P.* filter to retain comparability with much of *RBC* research. Statistically, given y_t for $t = 1, 2, \ldots, T$ the series can be decomposed in

[1] For the normalization we used yearly data on population aged between 15 and 65. ISTAT provides the series.

$y_t = c_t + \tau_t + \varepsilon_t$, the *H.P.* filter optimally extracts a trend which is stochastic but moves smoothly over time and is uncorrelated with the cyclical component by minimizing:

$$\min_{\tau}\left(\sum_{t=1}^{T}(y_t-\tau_t)^2+\lambda\sum_{t=2}^{T-1}\left[(\tau_{t+1}-\tau_t)-(\tau_t-\tau_{t-1})\right]^2\right) \tag{4.1}$$

Where T is the sample size and λ is a parameters that penalize the variability of the trend. Having quarterly data we chose a smoothing parameter of 1600 to isolate cycles of $4-6$ years following standard literature.[2]

4.1.2 Model output treatment

The model, as anticipated, has been loglinearized, and to simplify computation has been introduced in dynare already log-linearized wich means that the simulated variables are approximately[3] percentage deviations from the trend:

$$\hat{y} \equiv ln(y) - ln(\tilde{y}) \tag{4.2}$$

where in this case $\hat{y}$ represents a variable of interest. The output derived from the model has been *H.P.* filtered with a conventional smoothing parameter. At first sight, detrending a stationary model in order to compare moments (*i.e.* relative variances, covariances, autocorrelations) can seem useless since the model is, by definition, stationary. However as noted by Pfeifer (2014) it is a common practice to detrend model output to obey at the golden rule of treating data and simulated variables in the same way. This ensures:

- Moments are comparable. That means that the *frequency band* extracted in the data is comparable to the model.
- Possible distortions due to data treatment are symmetrical (Sims-Cogley-Nason-

[2] Even in this case some debate exists, to mention Nelson and C. R. Plosser (1982) that debate on the magnitude of λ.

[3] The fact that variables are approximately percentage deviations from the trend is trivially proved: $\hat{y} \equiv ln(y) - ln(\tilde{y}) = ln(\frac{y}{\tilde{y}}) = ln(1+\frac{dev}{\tilde{y}})$ that with the development of Taylor-McLaurin at the first grade is equal to $dev/\tilde{y}$ *i.e* percentage change.

approach[4]).

4.2 IRF analysis

We analyze IRF to a unit shock to productivity. IRF are really similar qualitatively to Gertler and Trigari (2009); we tend to mimic the same responses, but still one thing has to be noted, due to data availability we chose a quarterly calibration[5] which means that, given the same microeconomic evidence on duration of the contracts, our λ is greatly smaller than the one considered in Gertler and Trigari (2009). For reference, in *Table 2* we reported calibrated parameters for both the economies, our $\lambda = 0.75$ means that on average 25% of firms renegotiate in the next period, Gertler and Trigari (2009) calibrate $\lambda = 0.889$ which implies that in the next period only 11.1% of firms renegotiate the contracts. Having analyzed that our Poisson's adjustment creates much less stickiness in the model we can analyze the model's performance against the real economy.

4.3 Moment matching

We now confront three models and their performances in replicating labor market statistics. To be precise 3 models are considered: a non staggered Nash bargaining, one where the average duration before a renegotiation happens of 1 year and one of 2 years. Summary statistics are reported in *Table 3*. Overall the model performs well on autocorrelation and relative correlation, but fails to capture the relative standard deviation of the employed and unemployed. Gertler and Trigari (2009) model over performs baseline RBC without staggered Nash bargaining, especially on the labor market side. We suggests that the reason why Gertler and Trigari (2009) fails, has to do with the quarter calibration of

[4]See Christiano, Eichenbaum, et al. (2006).

[5]The choice was dictated, not by comparison purposes, since we aggregated dynare output and filtered to obtain and replicate the original findings of the paper, but was rather due to calibration. Since, as already recalled, much of the literature is on US, the idea to use already computed parameters and transform those in monthly parameters was not appealing. As example even with the permanent inventory approach, we were able only to derive a quarterly series for capital.

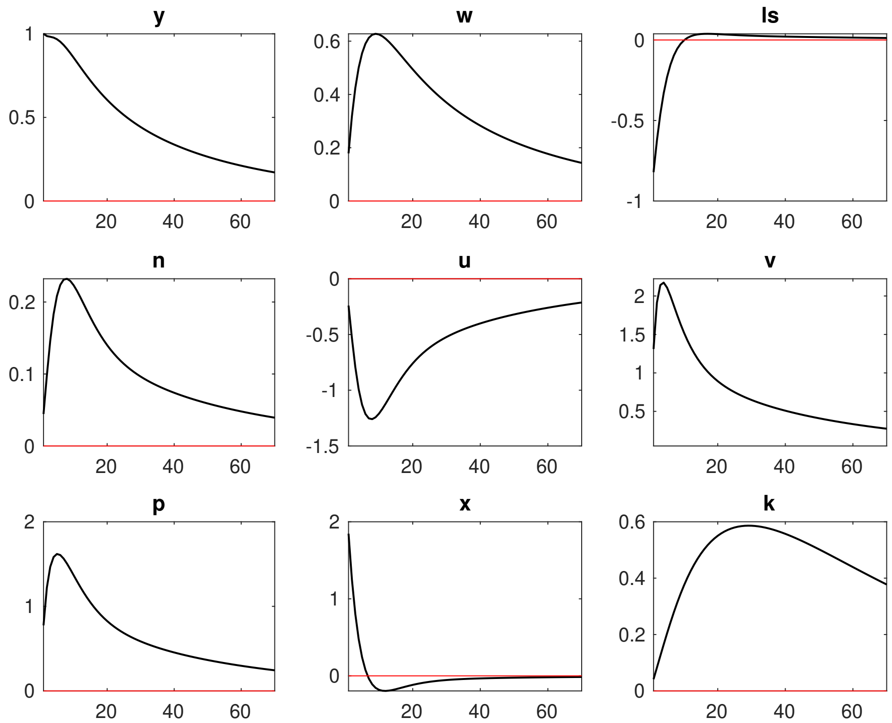

Figure 4.1: Impulse response function to a unit shock to productivity

the model. Ceteris paribus a quarterly calibrated model imply a λ that is far below the equivalent monthly calibration. This implies that the rigidity of the model is far less and that is why, the $8Q$ renegotiating period is performing better. It could well be that the renegotiation time is greater than one or two years, even though this seems unlikely. A more profound analysis in relation to the collective bargaining should be carried to distinguish between the two different explanations.

Table 3: Aggregate Statistics

	y	n	u	k
	Italy, 2004:1 - 2019:4			
Relative standard deviation	1.00	0.50	4.04	0.23
Autocorrelation	0.88	0.89	0.89	0.75
Correlation y	1.00	0.72	-0.74	0.46
	Model Economy, $\lambda \to 0Q$			
Relative standard deviation	1.00	0.12	0.53	0.17
Autocorrelation	0.73	0.96	0.96	0.96
Correlation y	1.00	0.38	-0.38	0.25
	Model Economy, $\lambda \to 4Q$			
Relative standard deviation	1.00	0.2	0.87	0.18
Autocorrelation	0.75	0.95	0.95	0.97
Correlation y	1.00	0.65	-0.65	0.19
	Model Economy, $\lambda \to 8Q$			
Relative standard deviation	1.00	0.32	1.37	0.18
Autocorrelation	0.76	0.96	0.96	0.97
Correlation y	1.00	0.64	-0.64	-0.13

4.4 Concluding Remarks

We have derived Gertler and Trigari (2009) model, we calibrated the model on Italy and judged it against Italian data. Even though the model fails to incorporate the large relative standard deviation of employment and unemployment, the model has an edge over the non staggered model. This could be due to a long average time before renegotiation happens or rather because of the quarterly calibration. It would be interesting to confront for the same period Maffezzoli (2001) model that incorporates specificity of Italian labor market such as collective bargaining and eventually developing a model with staggered collective bargaining to understand the magnitude of the two effects. This represent an interesting research program for the next years.

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Appendix A

Contraction mapping

This appendix sketches the mathematical background behind the Bellman equation, a very powerful instrument used in dynamic programming and in reinforcement learning. A small introduction of metric spaces, complete metric spaces, contractions as well as Blackwell's sufficient conditions follows in order to properly introduce the workhorse of dynamic programming: contraction mapping. The end of the section deals with Bellman equation as an application of the fixed point theorem.

A.1 Setting definitions

Definitions and theorems here are reported from the book "Recursive Macroeconomic theory" by Ljungqvist and Sargent (2004). Some definitions or theorems are contracted by using more formalism, clearly reducing the clarity of the exposition, with the intention of keeping small this part of the appendix which has the only scope of introducing the instruments to praise contraction mapping. For a more systematic approach referring to the book is suggested. Some personal interpretations, with the idea of helping visualize are given, warning to this interpretation and approach are contained in note[1].

Definition A.1.1. *A metric space is a set X and a function $d : X \times X \to R$ that satisfy four properties:*

A1. Positivity $d(x, y) \geq 0$ for all $x, y \in X$

A2. Strict positivity $d(x, y) = 0 \iff x = y$

A3. Symmetry $d(x, y) = d(y, x)$ for all $x, y \in X$

A4. Triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$

Finding examples of a metric space is quite easy since proving A1 - A3 is usually pretty trivial. A metric space can be interpreted as a set and something that, by abstraction, works as a distance. Clearly the metric can be different from the usual distance, but the "needed" properties of the distance at the end are contained in A1 - A4, that's why we can visualize, in an abstract environment, the metric as a sort of distance. To be more precise the metric is the abstraction of the common distance.

[1]No need to say the best way to communicate math is with formalism, but at times in the appendix there will be personal interpretations of the different objects. These interpretations should be considered for what they are, a way to increase the fluidity of the text and a way to help imagination. Not only interpretations are formally wrong but caution should be taken since narrating math is a double hedge sword with the major risk of trivializing the subject.

Definition A.1.2. *A metric space* (X, d) *is said to be complete if each Cauchy sequence in* (X, d) *is a convergent sequence in* (X, d)*. That is, in a complete metric space, each Cauchy sequence converges to a point belonging to the metric space.*

Recall that a necessary condition to converge is being a Cauchy series, so a complete metric space is a metric space in which every sequence that can converge, converge. This can be easily shown as follows:

Definition A.1.3. *A sequence* $\{x_n\}$ *in a metric space* (X, d) *is said to be a Cauchy sequence if for each* $\epsilon > 0$ *there exists an* $N(\epsilon)$ *such that* $d(x_n, x_m) < \epsilon \ \forall n, m \geq N(\epsilon)$*. Thus a sequence* $\{x_n\}$ *is said to be Cauchy if* $\lim_{n.m\to\infty} d(x_n, x_m) = 0$*.*

Definition A.1.4. *A sequence* $\{x_n\}$ *in a metric space* (X, d) *is said to converge to a limit* $x_0 \in X \ if \ \forall\epsilon > 0 \ \exists \ N(\epsilon) : d(x_n, x_0) < \epsilon \ for \ n \geq N(\epsilon)$*.*

Theorem A.1.5. *Let* $\{x_n\}$ *be a convergent sequence in a metric space* (X, d)*. Then* $\{x_n\}$ *is a Cauchy sequence.*

Proof. If $\epsilon > 0, x_0 \in X : x_0 = limit \ \{x_n\} \ then \ \forall m, n \ d(x_n, x_m) \leq d(x_n, x_0) + d(x_m, x_0)$ Since x_0 is the limit of the series, $\exists N : d(x_n, x_0) < \frac{\epsilon}{2} \ \forall n \geq N$, this two inequalities imply that $d(x_n, x_m) < \epsilon \ \forall \ n, m \geq N$, so the series is Cauchy. □

Let's now introduce a couple of definitions that will be needed in the future: operator and continuity of an operator. The idea behind an operator is quite simple, an operator is a function from an X subset to the same X subset. More will come with the explanation of contraction mapping.

Definition A.1.6. *A function* f *mapping a metric space* (X, d) *into itself is called an operator.*

Definition A.1.7. *Let* f *be an operator in the metric space* (X, d)*. The operator is said to be continuous at the point* $x_0 \in X$ *if* $\forall\epsilon > 0 \ \exists \ \delta > 0 : \ d\,[f(x), f(x_0)] < \epsilon$ *whenever* $d(x, x_0) < \delta$*.*

A.2 Contraction Mapping

Definition A.2.1. *Let (X, d) be a metric space and let $f : X \to X$. We say that f is a contraction or contraction mapping if there is a real number $k, 0 \leq k < 1$, such that:*

$$d\left[f(x), f(y)\right] \leq kd(x, y)\ \forall x, y \in X.$$

Theorem A.2.2. *Let (X, d) be a complete metric space and let $f : X \to X$ be a contraction. Then there is a unique point $x_0 \in X\ \ f(x_0) = x_0$. Furthermore, if x is any point in X and $\{x_n\}$ is defined inductively according to $x_1 = f(x), x_2 = f(x1), ..., x_{n+1} = f(x_n)$, then $\{x_n\}$ converges to x_0.*

This is the famous *Banach fixed point theorem* also known as *contraction mapping theorem*, a great result both from the mathematical standpoint and the dynamic programming standpoint. For the proof, which is not complex refer to the book. The idea is simple, suppose having a metric space and a function from the complete metric space to itself, if this function is a contraction (and intuitively *nomen omen*, this function is indeed acting like a contraction) then there is always a fixed point *i.e. a non moving point.* What is really interesting and will be useful to explain the Bellman equation is that it is possible to find this point, which surely exists, with a recursive method. Heuristically:

$$f(f^{\infty}(x_0)) = f^{\infty+1}(x_0) = f^{\infty}(x_0)$$

This explanation, which is more than heuristic[2] gives an idea of how the recursive procedures work. In this appendix the attention is on dynamic programming application but the fixed point theorem is such a great theorem because it can be applied to a lot of different problems, as challenge the reader can try to think at the Newton's method as an application of the fixed point theorem.

It needs to be noted that the set X has't been defined, so it can possibly be a class of functions i.e. *space of function $Y \to K$*. To define a metric space we should add a function $f : X \times X \to R$ that is respecting A1 - A4, one candidate is the sup metric

[2]But made use of a property that each student wants to write down.

$d_\infty(x, y) = sup_t |x(t) - y(t)|$. To have some sense this particular "distance" that has been defined the class of function should be a class of bounded functions: let $X \subseteq R^l$ *then* $C(x)$ is the class of bounded functions $f : X \to R$. Note that the boundedness of the function is necessary because otherwise the sup metric can be non existent.

Theorem A.2.3 (Blackwell's Sufficient Conditions for T to be a Contraction)**.** *Consider a space of function X with the constraints previously imposed, let T be an operator defined on the metric* (X, d_∞). *Let T satisfy two properties:*

1. *Monotonicity:* $\forall x, y \in X,\ x \geq y$ *implies* $T(x) \geq T(y)$.

2. *Discounting: Let c denote a function that is constant at the real value c for all points in the domain of definition of the functions in X. For any positive real c and every* $x \in X,\ T(x + c) \leq T(x) + \beta c$ *for some* β *satisfying* $0 \leq \beta < 1$.

Then T *is a contraction mapping with modulus* β .

Again the proof will be eluded in order to finally present the Bellman equation as an application of contraction mapping.

A.3 From Banach to Bellman

Consider a problem such as:

$$v(X) = max_{u \in R^k}\{r(x,u) + \beta v(x^{'})\}, \quad x^{'} \leq g(x,u), \;\; 0 < \beta < 1 \tag{A.1}$$

We shall now assume that $r(x,u)$ is real valued, continuous, concave and bounded and that the set $\left[x^{'}, x, u \;\; : x^{'} \leq g(x,u), \;\; u \in R^k\right]$ is convex and compact. In addition consider the operator:

$$Tv = max_{u \in R^k}\{r(x,u) + \beta v(x^{'})\}, \quad x^{'} \leq g(x,u), \;\; x \in X$$

It can be shown (that's why we need that particular set of assumption on the return function and on the set) that the operator T map continuous bounded functions in continuous bounded functions. We are surely working in a metric space since we are considering a set X and a metric on set X that is respecting properties A1 - A4. Then we can apply the *Banach fixed point theorem* and so we ensure that the problem $v = Tv$ has always a unique solution since the operator T respect the monotonicity and discounting properties and so we can apply the *Blackwell sufficient conditions.* The fixed point is approached in the d_∞ metric by iterations $v^n = T^n(v_0)$ starting from any bounded continuous function v_0 . Convergence on the sup metric implies uniform convergence of the function. To recap what has been done in this appendix, we showed that the solution to the bellman equation (the value function) is a fixed point of a contraction with operator T and this implies that there is always a fixed point (in this case a value function solving the Bellman equation) and that we can find this function by starting with any bounded function and then iterating on this function the Bellman operator. Formally this is the most elegant solution, but it's not always the most efficient way to procede. In our model we solved the Bellman equation analytically with the Benveniste-Scheinkman equation eluding finding the value function. Other algorithms to solve the Bellman equation have been proposed such as the *Howard improvement.*

Appendix B

Benveniste and Scheinkman

Suppose that the problem is set as:

$$\max_u \{r(x,u) + \beta V(\tilde{x})\} \tag{B.1}$$

Under the constraint that $\tilde{x} = g(x,u)$ given x. Under some assumptions on r and g[1], Benveniste and Scheinkman (1979) show that the value function is differentiable, with derivative equal to:

$$V^{'}(x) = \frac{\delta r}{\delta x}\left[x, h(x)\right] + \beta \frac{\delta g}{\delta x}\left[x, h(x)\right] V^{'}\{g\left[x, h(x)\right]\} \tag{B.2}$$

If, as we did in the text, the transition law do not include the state x, then (B.2) becomes a simple *Envelope Theorem*:

$$V^{'}(x) = \frac{\delta r}{\delta x}\left[x, h(x)\right] \tag{B.3}$$

[1]For the description of the assumptions we redirect to the original paper Benveniste and Scheinkman (1979)